GRADUATE LECTURE NOTES • SCUOLA SUPERIORE SANT'ANNA

The Integrate-and-Fire Neuron

Lecture 1 — From the membrane equation to spiking dynamics and extensions

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Abstract

We build the integrate-and-fire (IF) neuron from first principles, with every step of the derivations worked out, following the treatment of Gerstner, Kistler, Naud & Paninski, *Neuronal Dynamics* (EPFL) [1]. We derive the membrane equation from ionic biophysics, solve the leaky integrate-and-fire (LIF) dynamics in full (integrating factor, Dirac Green's function, harmonic impedance), obtain the frequency–current relation by explicit inversion, and treat the noisy neuron through its numerical simulation (the Euler–Maruyama scheme for stochastic differential equations). We close with a detailed study of the nonlinear and adaptive extensions — spike-frequency adaptation, the adaptive exponential model, and the Izhikevich model and its firing regimes.

Contents

1 Biophysical foundations

1.1 The Nernst potential, derived

A membrane separates two solutions of an ion s (valence z_s) at concentrations $[s]_{\text{in}}, [s]_{\text{out}}$. The molar flux density combines Fick diffusion and electric drift (the Nernst–Planck equation):

$$J_s = -D_s \frac{\mathrm{d}c_s}{\mathrm{d}x} - \frac{z_s F}{RT} D_s\, c_s \frac{\mathrm{d}\phi}{\mathrm{d}x}, \tag{1}$$

with D_s the diffusivity, $\phi(x)$ the electric potential, F Faraday's constant, R the gas constant, T temperature. *Step 1 — equilibrium.* At electrochemical equilibrium the net flux vanishes, $J_s = 0$, so

$$D_s \frac{\mathrm{d}c_s}{\mathrm{d}x} = -\frac{z_s F}{RT} D_s\, c_s \frac{\mathrm{d}\phi}{\mathrm{d}x} \implies \frac{1}{c_s}\frac{\mathrm{d}c_s}{\mathrm{d}x} = -\frac{z_s F}{RT}\frac{\mathrm{d}\phi}{\mathrm{d}x}. \tag{2}$$

Step 2 — separate and integrate across the membrane (from inside to outside):

$$\int_{[s]_{\text{in}}}^{[s]_{\text{out}}} \frac{\mathrm{d}c_s}{c_s} = -\frac{z_s F}{RT}\int_{\phi_{\text{in}}}^{\phi_{\text{out}}} \mathrm{d}\phi \implies \ln\frac{[s]_{\text{out}}}{[s]_{\text{in}}} = -\frac{z_s F}{RT}\big(\phi_{\text{out}} - \phi_{\text{in}}\big). \tag{3}$$

Step 3 — solve for the equilibrium membrane potential $E_s \equiv \phi_{\text{in}} - \phi_{\text{out}}$:

$$\boxed{E_s = \frac{RT}{z_s F}\ln\frac{[s]_{\text{out}}}{[s]_{\text{in}}}.} \tag{4}$$

At 37°C, $RT/F \approx 26.7\,\text{mV}$, giving $E_{\text{Na}} \approx +60$, $E_{\text{K}} \approx -90$, $E_{\text{Cl}} \approx -65\,\text{mV}$.

1.2 Leak, capacitance, and current balance

Near rest the many background conductances combine into a single Ohmic *leak* $I_L = g_L(V - E_L)$ with $E_L \approx V_{\text{rest}}$ [1, 3]. The bilayer stores charge $Q = C_m V$; differentiating gives the capacitive current $I_C = \mathrm{d}Q/\mathrm{d}t = C_m\,\mathrm{d}V/\mathrm{d}t$. Treating the soma as isopotential, charge conservation at the node (Kirchhoff) reads

$$C_m \frac{\mathrm{d}V}{\mathrm{d}t} = -I_{\text{ion}}(V,t) + I(t). \tag{5}$$

1.3 From Hodgkin–Huxley to integrate-and-fire

The full biophysics of spike generation is the Hodgkin–Huxley (HH) model, in which the ionic current is the sum of voltage-gated sodium, potassium, and leak currents:

$$C_m \dot{V} = -\bar{g}_{\text{Na}} m^3 h (V - E_{\text{Na}}) - \bar{g}_{\text{K}} n^4 (V - E_{\text{K}}) - g_L (V - E_L) + I(t), \tag{6}$$

with gating variables $x \in \{m, h, n\}$ each obeying $\dot{x} = \alpha_x(V)(1-x) - \beta_x(V)\,x$. Sodium activation m is fast and regenerative: past a threshold it opens explosively, producing the upstroke; then inactivation h and the potassium variable n repolarize and transiently hyperpolarize the cell (the refractory period). Two facts license the IF reduction: (i) the spike is *stereotyped*, so its waveform carries little information and may be replaced by an event plus a reset; (ii) *between* spikes the dynamics are nearly linear, set by the leak. Discarding the spike shape and keeping the subthreshold leak yields exactly the LIF equation below with a threshold and reset — the IF neuron is the analytic *caricature* of HH. The exponential and quadratic models of §7 restore an approximation of the sodium upstroke.

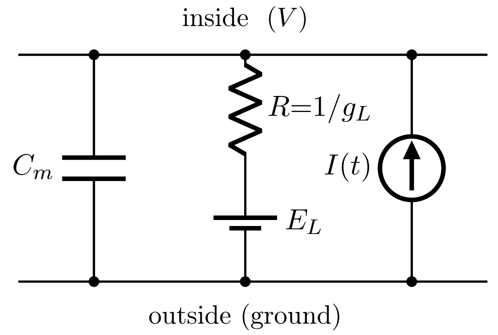

Figure 1: Equivalent circuit of an isopotential membrane patch (cf. [1], Ch. 1). The injected current $I(t)$ points *into* the intracellular node; the leak branch is $R = 1/g_L$ in series with the reversal battery E_L.

1.4 The cable equation and the isopotential assumption

Real neurons are spatially extended. Treating a dendrite as a thin cable of radius a and axial resistivity r_i, current conservation gives the **cable equation**

$$c_m\,\partial_t V = \frac{a}{2r_i}\,\partial_x^2 V - i_{\text{ion}}(V) + i_{\text{ext}}(x,t), \tag{7}$$

a reaction–diffusion PDE whose electrotonic length constant $\lambda = \sqrt{aR_m/2r_i}$ sets how far voltage spreads. When the cell is small compared with λ (or we lump it into one compartment) the diffusion term $\partial_x^2 V$ is negligible and the PDE collapses to the single ODE (5): this *isopotential* assumption is what we use throughout. Multi-compartment models reinstate $\partial_x^2 V$ to capture dendritic processing.

2 The leaky integrate-and-fire model

Keeping only the leak, $I_{\text{ion}} = g_L(V - E_L)$, equation (5) reads $C_m\dot{V} = -g_L(V-E_L) + I(t)$. *Dividing through by* g_L,

$$\frac{C_m}{g_L}\frac{\mathrm{d}V}{\mathrm{d}t} = -(V - E_L) + \frac{I(t)}{g_L}, \tag{8}$$

and *defining* the membrane time constant $\tau_m \coloneqq C_m/g_L$ and resistance $R \coloneqq 1/g_L$ (so $\tau_m = RC_m$),

$$\boxed{\tau_m\frac{\mathrm{d}V}{\mathrm{d}t} = -(V - E_L) + R\,I(t).} \tag{9}$$

This passive integrator cannot spike; we add the **fire-and-reset** rule

$$V(t^-) = \vartheta \;\Rightarrow\; \text{spike at } t, \quad V(t^+) = V_r, \tag{10}$$

(threshold ϑ, reset V_r, optional refractory τ_{ref}). The LIF is a *hybrid* system: smooth linear flow (9) interrupted by the discrete map (10) (Fig. 2).

How to read Fig. 2 (worked walk-through). Before the current is injected the cell sits at rest, $V = E_L = V_r$ (flat line). When the step turns on, the right-hand side of (9) becomes positive, so $\mathrm{d}V/\mathrm{d}t > 0$ and V rises. It would relax exponentially toward the *asymptote* $E_L + RI_0$ (the value at which leak and input currents balance), but here that asymptote (1.5) lies *above* the threshold (1): the potential never gets there. Instead, when V reaches ϑ the fire-and-reset rule (10) triggers — we mark a spike and instantaneously set $V \to V_r$ — and the charging starts over. Because the drive is constant, every cycle is identical, so the neuron fires *periodically* with the

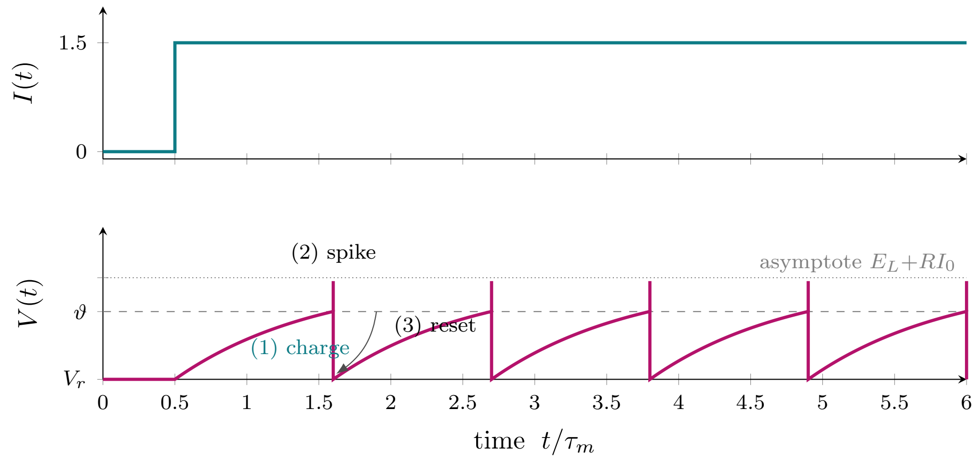

Figure 2: LIF response to a constant step current. *Top*: $I(t)$ switched on at $t = 0.5\,\tau_m$. *Bottom*: the membrane potential (1) *charges* exponentially toward the asymptote $E_L + RI_0$ (dotted), (2) hits the threshold ϑ (dashed) and *fires* a spike (vertical mark), then (3) *resets* to V_r; the cycle repeats with period $T = \tau_m \ln \frac{E_L+RI_0-V_r}{E_L+RI_0-\vartheta}$. Cf. [1], §1.3.

interval T derived in (22). Two limits are worth noting for students: if the asymptote were *below* threshold ($E_L + RI_0 < \vartheta$) the neuron would charge up, stall below ϑ, and never fire (sub-rheobase); the closer the asymptote sits to ϑ, the slower the final approach and the longer T — this is the origin of the arbitrarily low firing rates near rheobase.

A geometric picture: the phase line. A second way to see the same dynamics is to plot the *rate of change* $\mathrm{d}V/\mathrm{d}t$ against V itself (Fig. 3). For the LIF this is the straight line $\mathrm{d}V/\mathrm{d}t = [\,(E_L + RI_0) - V\,]/\tau_m$: positive when V is below the asymptote (so V moves right) and zero at the asymptote. The flow therefore pushes V rightward, but the threshold ϑ acts as a trap-door before the asymptote is reached: upon contact, V is teleported back to V_r and the motion repeats. This phase-line view is what generalizes to the nonlinear models of §7, where the straight line becomes curved and can create or destroy fixed points.

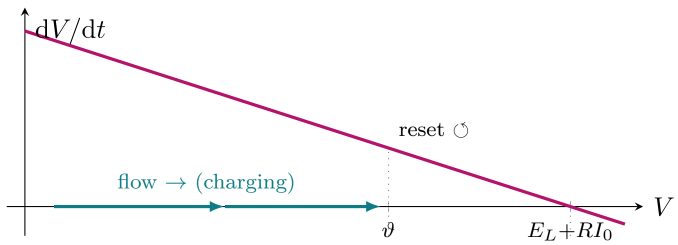

Figure 3: Phase line of the LIF: $\mathrm{d}V/\mathrm{d}t$ versus V. The flow (arrows) carries V toward the asymptote $E_L + RI_0$ where $\mathrm{d}V/\mathrm{d}t = 0$, but the threshold ϑ intercepts it first and resets it to V_r (curved arrow). When $E_L + RI_0 < \vartheta$ the line crosses zero before ϑ and the neuron settles silently (sub-rheobase).

3 Subthreshold dynamics, derived in full

3.1 General solution by the integrating factor

Rewrite (9) in standard linear form by dividing by τ_m:

$$\frac{\mathrm{d}V}{\mathrm{d}t} + \frac{1}{\tau_m} V = \frac{1}{\tau_m}\big(E_L + RI(t)\big). \tag{11}$$

Step 1 — integrating factor. Multiply both sides by $\mu(t) = e^{t/\tau_m}$. The left side becomes an exact derivative because $\frac{\mathrm{d}}{\mathrm{d}t}\big(e^{t/\tau_m} V\big) = e^{t/\tau_m}\dot{V} + \frac{1}{\tau_m} e^{t/\tau_m} V$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(e^{t/\tau_m} V\Big) = \frac{1}{\tau_m} e^{t/\tau_m}\big(E_L + RI(t)\big). \tag{12}$$

Step 2 — integrate from t_0 to t:

$$e^{t/\tau_m} V(t) - e^{t_0/\tau_m} V(t_0) = \frac{1}{\tau_m}\int_{t_0}^{t} e^{s/\tau_m}\big(E_L + RI(s)\big)\,\mathrm{d}s. \tag{13}$$

Step 3 — evaluate the E_L piece: $\frac{E_L}{\tau_m}\int_{t_0}^{t} e^{s/\tau_m}\mathrm{d}s = E_L\big(e^{t/\tau_m} - e^{t_0/\tau_m}\big)$. *Step 4 — divide by* e^{t/τ_m} and use $R/\tau_m = 1/C_m$:

$$\boxed{V(t) = E_L + \big(V(t_0) - E_L\big)e^{-(t-t_0)/\tau_m} + \frac{1}{C_m}\int_{t_0}^{t} e^{-(t-s)/\tau_m} I(s)\,\mathrm{d}s.} \tag{14}$$

3.2 Dirac impulse and the Green's function

Take the input to be a charge q delivered instantaneously, $I(t) = q\,\delta(t)$, with the cell at rest before $t = 0$ ($t_0 \to -\infty$, $V(t_0) = E_L$). The convolution term in (14) is evaluated with the *sifting property* $\int f(s)\delta(s-a)\mathrm{d}s = f(a)$:

$$V(t) - E_L = \frac{1}{C_m}\int_{-\infty}^{t} e^{-(t-s)/\tau_m}\, q\,\delta(s)\,\mathrm{d}s = \frac{q}{C_m}\, e^{-t/\tau_m} \quad (t > 0). \tag{15}$$

Hence the potential *jumps* by q/C_m at $t = 0$ and relaxes exponentially. Writing $V(t) - E_L = q\,\kappa(t)$ identifies the **Green's function** (impulse response / elementary PSP)

$$\kappa(t) = \frac{1}{C_m}\, e^{-t/\tau_m}\,\Theta(t), \tag{16}$$

and by linearity the response to *any* input is the convolution

$$V(t) = E_L + (\kappa * I)(t) = E_L + \int_{-\infty}^{t} \kappa(t-s) I(s)\,\mathrm{d}s : \tag{17}$$

the LIF membrane is a first-order low-pass filter.

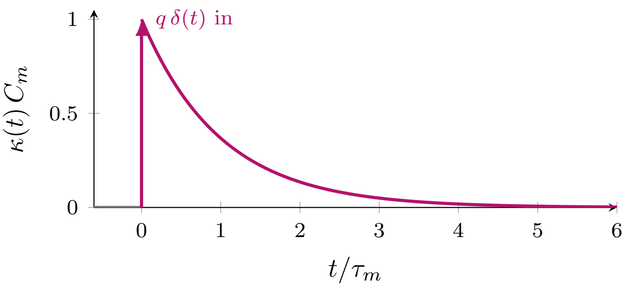

Figure 4: Green's function (16): response to a Dirac impulse $q\,\delta(t)$ — a jump followed by e^{-t/τ_m} decay (the elementary postsynaptic potential).

Temporal summation: integrating inputs. Because the membrane is linear (17), the response to a train of synaptic pulses is the *sum* of the individual postsynaptic potentials of Fig. 4. This is the "integrate" in integrate-and-fire. Figure 5 simulates a neuron receiving brief inputs (arrows): *isolated* inputs (at $t = 1, 3$) each produce a PSP that simply decays away, but a *rapid burst* of inputs arriving within roughly one time constant *summate* — each new PSP rides on the tail of the previous one — until the accumulated depolarization crosses ϑ and the neuron fires. The neuron thus acts as a leaky coincidence detector: inputs that are clustered in time are far more effective than the same inputs spread out, because the leak forgets the early ones.

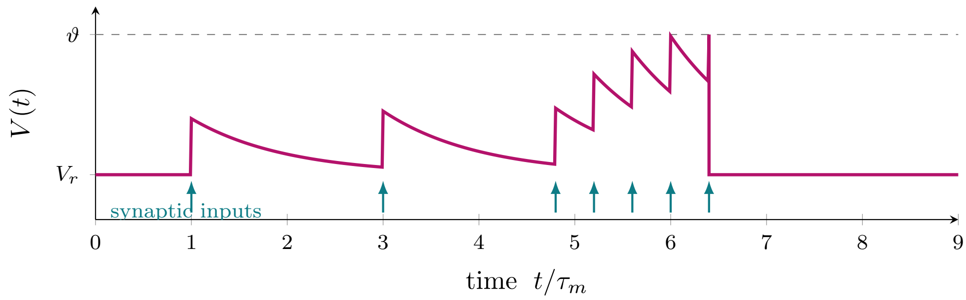

Figure 5: Temporal summation (simulated LIF with delta inputs, $\tau_m = 1$). Isolated inputs decay without effect; the fast cluster near $t \approx 5$–6 summates past ϑ and triggers a spike, after which V resets. The leak makes the neuron sensitive to input *timing*.

3.3 Step response, derived

For a current step $I(t) = I_0\,\Theta(t)$ with $V(0) = V_0$, evaluate the convolution integral in (14) explicitly:

$$\begin{aligned}\frac{1}{C_m}\int_0^t e^{-(t-s)/\tau_m} I_0\,\mathrm{d}s &= \frac{I_0}{C_m}\,e^{-t/\tau_m}\int_0^t e^{s/\tau_m}\mathrm{d}s = \frac{I_0}{C_m}\,e^{-t/\tau_m}\,\tau_m\big(e^{t/\tau_m}-1\big)\\ &= \frac{I_0\tau_m}{C_m}\big(1-e^{-t/\tau_m}\big) = RI_0\big(1-e^{-t/\tau_m}\big),\end{aligned} \tag{18}$$

using $\tau_m/C_m = R$. Therefore

$$V(t) = V_\infty + (V_0 - V_\infty)\,e^{-t/\tau_m}, \qquad V_\infty = E_L + RI_0, \tag{19}$$

the smooth charging curve of Fig. 2.

3.4 Harmonic response: impedance, derived

Seek the stationary response to $I(t) = I_0 e^{i\omega t}$ with the ansatz $V(t) - E_L = Z(\omega) I_0 e^{i\omega t}$. Substitute into (9) (in the form $\tau_m \dot{V} + (V - E_L) = RI$):

$$\tau_m\,(i\omega) Z I_0 e^{i\omega t} + Z I_0 e^{i\omega t} = R\,I_0 e^{i\omega t} \implies Z(\omega)\,(1 + i\omega\tau_m) = R, \tag{20}$$

so $Z(\omega) = R/(1 + i\omega\tau_m)$. *Rationalize* (multiply by the conjugate) to read off modulus and phase:

$$Z(\omega) = \frac{R(1 - i\omega\tau_m)}{1 + (\omega\tau_m)^2}, \qquad |Z(\omega)| = \frac{R}{\sqrt{1 + (\omega\tau_m)^2}}, \qquad \arg Z(\omega) = -\arctan(\omega\tau_m). \tag{21}$$

The amplitude is flat below the cutoff $f_c = 1/(2\pi\tau_m)$ and falls as $1/\omega$ above it; the phase lags from $0°$ to $-90°$ (Fig. 6). The passive LIF has no resonance.

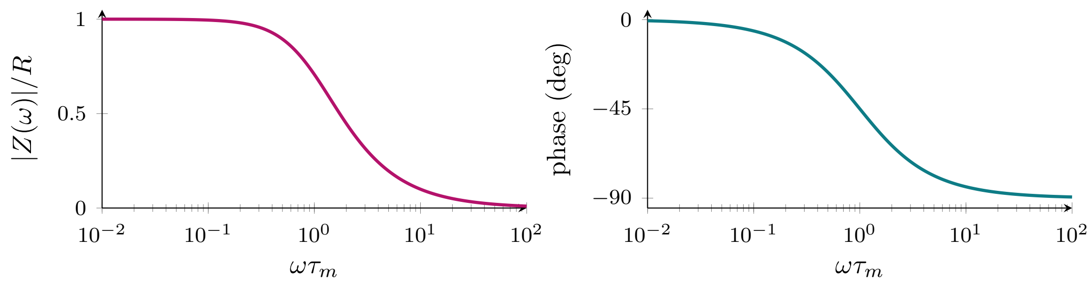

Figure 6: Bode diagram of the membrane filter (21): amplitude (left) and phase (right).

4 Deterministic firing and the f–I curve, derived

With constant supra-threshold drive the trajectory between resets starts at $V(0) = V_r$ and follows (19), $V(t) = V_\infty + (V_r - V_\infty)e^{-t/\tau_m}$. *Impose the threshold condition* $V(T) = \vartheta$ and solve for the interval T:

$$\vartheta = V_\infty + (V_r - V_\infty)e^{-T/\tau_m} \implies e^{-T/\tau_m} = \frac{\vartheta - V_\infty}{V_r - V_\infty} = \frac{V_\infty - \vartheta}{V_\infty - V_r},$$
$$T = \tau_m \ln \frac{V_\infty - V_r}{V_\infty - \vartheta} = \tau_m \ln \frac{RI_0 + E_L - V_r}{RI_0 + E_L - \vartheta}. \tag{22}$$

Adding the refractory period, the rate is $r = 1/(\tau_{\text{ref}} + T)$:

$$\boxed{r(I_0) = \left[\tau_{\text{ref}} + \tau_m \ln \frac{RI_0 + E_L - V_r}{RI_0 + E_L - \vartheta}\right]^{-1}, \quad I_0 > I_{\text{rh}}.} \tag{23}$$

Rheobase: the argument of the logarithm diverges (so $T \to \infty$, $r \to 0$) when $RI_0 + E_L = \vartheta$, i.e. at $I_{\text{rh}} = (\vartheta - E_L)/R$. *Strong drive*: expanding $\ln \frac{V_\infty - V_r}{V_\infty - \vartheta} = \ln\left(1 + \frac{\vartheta - V_r}{V_\infty - \vartheta}\right) \approx \frac{\vartheta - V_r}{V_\infty}$ for $V_\infty \gg \vartheta$ gives $r \approx RI_0/[\tau_m(\vartheta - V_r)]$, saturating at $1/\tau_{\text{ref}}$.

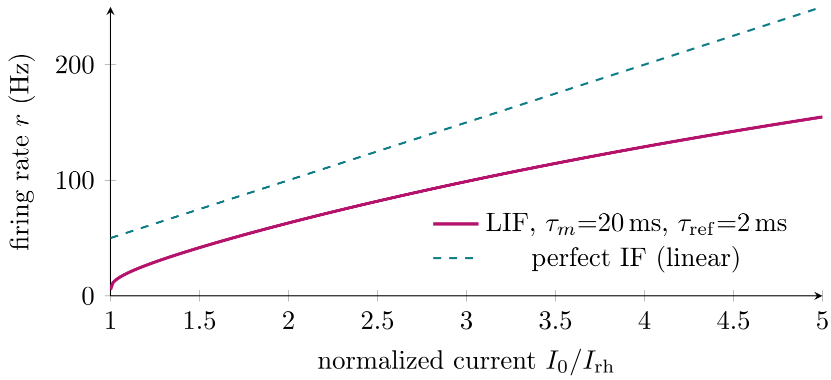

Figure 7: The f–I relation (23): continuous onset at rheobase, saturation toward $1/\tau_{\text{ref}}$ (solid); perfect IF is linear (dashed). Cf. [1, 4].

5 Spike trains as Dirac combs

A spike train is fully specified by its firing times $\{t_f\}$ through the spike-train density

$$\rho(t) = \sum_f \delta(t - t_f), \tag{24}$$

whose expectation is the instantaneous rate $\nu(t) = \langle \rho(t) \rangle$ and whose integral over a window is the spike count. Synaptic input is this comb filtered by a synaptic kernel α: $I_{\text{syn}} = q(\alpha * \rho)$, so by (17) the voltage is a sum of postsynaptic potentials $V(t) - E_L = q \sum_f (\kappa * \alpha)(t - t_f)$, the spike-response picture of §7.6.

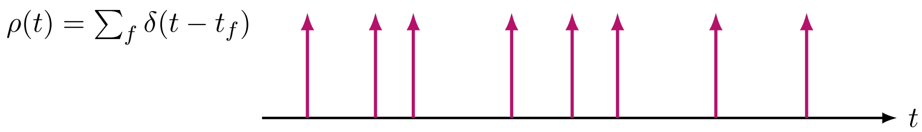

Figure 8: A spike train as a sum of Dirac deltas (a Dirac comb); each arrow marks a firing time.

Under constant drive the LIF fires periodically ($C_V = 0$), unlike cortical neurons ($C_V \approx 1$): irregularity comes from input fluctuations, derived next [1, 5].

6 The noisy neuron and its numerical simulation

6.1 From synaptic bombardment to a stochastic equation

A cortical neuron receives thousands of synaptic inputs. Modelling their net effect as a mean drive μ plus Gaussian fluctuations of intensity σ (justified when inputs are many, small, and fast), the LIF equation (9) becomes the **Langevin equation**

$$\tau_m \, \mathrm{d}V = \big[-(V - E_L) + \mu\big]\mathrm{d}t + \sigma\sqrt{\tau_m}\,\mathrm{d}W_t, \tag{25}$$

where W_t is a Wiener process (Brownian motion: $\mathrm{d}W_t$ is Gaussian, zero-mean, with $\langle \mathrm{d}W_t^2\rangle = \mathrm{d}t$), together with the reset rule (10). This is a *stochastic differential equation* (SDE): we cannot integrate it by hand for each realization, so we simulate it.

6.2 Numerical integration: the Euler–Maruyama method

The standard scheme for an SDE is **Euler–Maruyama**, the stochastic analogue of forward Euler. Discretize time into steps Δt, write $V_k \approx V(k\Delta t)$, and update

$$\boxed{V_{k+1} = V_k + \frac{1}{\tau_m}\big[-(V_k - E_L) + \mu\big]\Delta t \; + \; \frac{\sigma}{\sqrt{\tau_m}}\sqrt{\Delta t}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0,1) \text{ i.i.d.}} \tag{26}$$

The one feature that distinguishes this from a deterministic Euler step is that **the noise term scales as $\sqrt{\Delta t}$, not Δt**. The reason: over a step the Wiener increment $\Delta W = W_{k+1} - W_k$ is Gaussian with variance Δt, so $\Delta W = \sqrt{\Delta t}\,\xi_k$. (Writing Δt instead of $\sqrt{\Delta t}$ is a common bug — it makes the noise vanish as $\Delta t \to 0$.) The spike rule is checked after each update:

Algorithm (noisy LIF by Euler–Maruyama).
choose $\Delta t \ll \tau_m$; set $V_0 = V_r$, spike list empty.
for $k = 0, 1, 2, \ldots$:
draw $\xi_k \sim \mathcal{N}(0,1)$ and update V_{k+1} via (26);
if $V_{k+1} \geq \vartheta$: record a spike at $t = (k+1)\Delta t$ and set $V_{k+1} \leftarrow V_r$.

Euler–Maruyama has *strong* order $\frac{1}{2}$ and *weak* order 1 in Δt; in practice take Δt a small fraction of τ_m (e.g. $\Delta t \leq \tau_m/50$) and confirm the results do not change when Δt is halved. Every quantity of interest is then *measured* from the simulated trajectories rather than computed from any analytic formula: the firing rate as (number of spikes)/(time), the C_V from the inter-spike intervals, and the membrane-potential distribution as a histogram of the subthreshold V_k (Figs. 9–11). This Monte-Carlo recipe needs no heavy probability machinery and applies unchanged to the nonlinear models below, for which closed-form firing rates do not exist.

The key qualitative result. With a *subthreshold* mean drive ($\mu < \vartheta - E_L$) the deterministic neuron is silent, but the fluctuations occasionally push V across ϑ, producing *irregular* firing ($C_V \approx 1$). Noise therefore smooths the f–I curve and abolishes the hard rheobase — the *fluctuation-driven* regime in which real cortical and subcortical neurons operate.

The membrane-potential distribution. Histogramming the subthreshold voltage V_k from a long Euler–Maruyama run gives the membrane-potential distribution $p_0(V)$ (Fig. 9): it peaks near the mean drive and falls to zero at the threshold ϑ, because every trajectory that reaches ϑ is reset to V_r. How often that happens is exactly the firing rate — which here we simply count from the simulation.

What the noise does, in pictures. Figure 10 is a single Euler–Maruyama trajectory with subthreshold mean drive: the deterministic neuron would be silent, yet fluctuations push V across ϑ at irregular times. Counting spikes as the mean drive is varied gives the f–I curve of Fig. 11: the deterministic curve has a sharp rheobase corner, while noise rounds it off and yields a nonzero rate even below threshold.

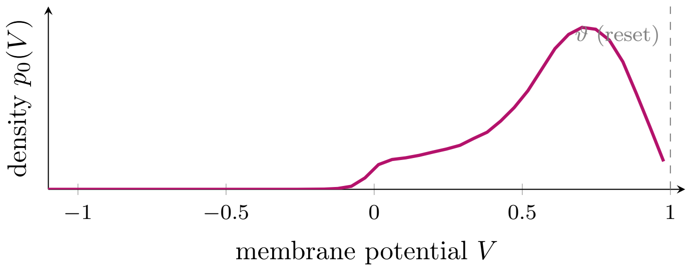

Figure 9: Membrane-potential distribution under noise, as a histogram of the subthreshold voltage from a long Euler–Maruyama simulation of (25). It peaks near the mean drive and vanishes at ϑ (every trajectory reaching threshold is reset to V_r).

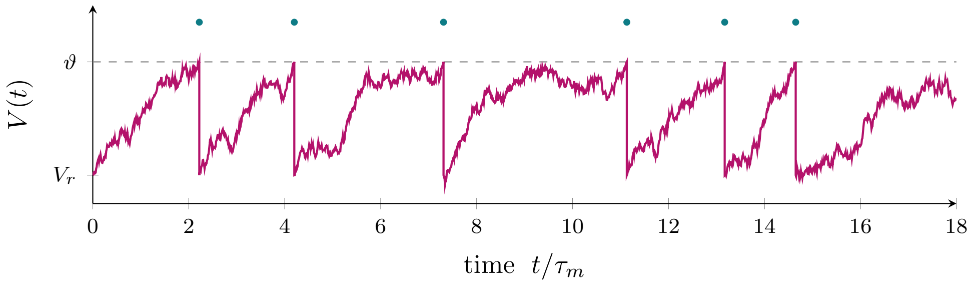

Figure 10: Fluctuation-driven firing (Euler–Maruyama simulation of (25), mean drive $\mu = 0.92\,\vartheta$, below rheobase). Subthreshold fluctuations occasionally cross ϑ (dots), giving *irregular* spikes ($C_V \approx 1$), unlike the clockwork firing of Fig. 2.

7 Nonlinear and adaptive extensions

7.1 Quadratic IF and the theta neuron

Expanding the spike nonlinearity to quadratic order gives $\tau_m \dot V = a(V-V_r)(V-V_c)+RI$, the normal form of a saddle-node-on-invariant-circle (SNIC) bifurcation. The substitution $V = \tan(\theta/2)$ (so $\dot V = \frac{1}{2}\sec^2(\theta/2)\,\dot\theta$ and $1+V^2 = \sec^2(\theta/2)$) maps it to the **theta neuron** $\dot\theta = (1-\cos\theta)+(1+\cos\theta)\,I$ on the circle — the canonical Type-I oscillator with $r \propto \sqrt{I - I_c}$ near onset.

7.2 Exponential integrate-and-fire (EIF)

Replacing the hard threshold by a spike-generating exponential (matching sodium-channel activation [7]):

$$\tau_m \dot V = -(V - E_L) + \Delta_T\, e^{(V-V_T)/\Delta_T} + RI(t), \tag{27}$$

recovers the LIF as $\Delta_T \to 0$ and produces a true (finite-time) spike upward divergence.

7.3 Adaptive exponential IF (AdEx): fixed points and stability, derived

Coupling (27) to a slow adaptation current w [6]:

$$C_m \dot V = \underbrace{-g_L(V - E_L) + g_L\Delta_T e^{(V-V_T)/\Delta_T} - w + I}_{\equiv F(V,w)}, \tag{28}$$

$$\tau_w \dot w = \underbrace{a(V - E_L) - w}_{\equiv G(V,w)}, \tag{29}$$

with reset $V \to V_r$, $w \to w + b$. *Fixed points* solve $F = G = 0$: the V-nullcline is the N-shaped curve $w = -g_L(V - E_L) + g_L\Delta_T e^{(V-V_T)/\Delta_T} + I$ and the w-nullcline the line $w = a(V - E_L)$; their

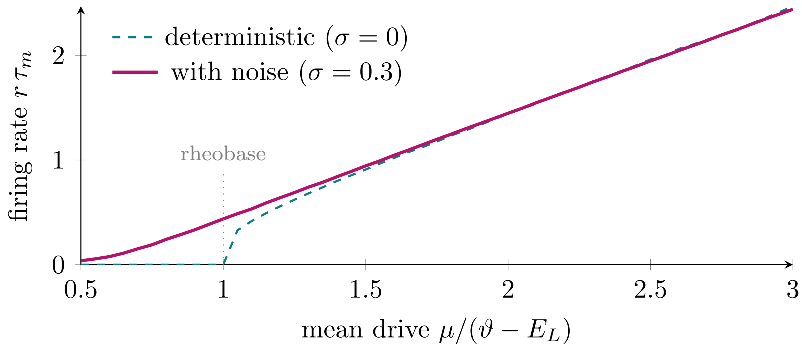

Figure 11: Noise smooths the f–I curve, here *measured by Euler–Maruyama Monte-Carlo* (spikes counted over a long simulation at each μ). The deterministic curve (dashed) is exactly zero below the rheobase ($\mu = \vartheta$) and rises with a sharp corner; noise (solid) rounds the corner and gives a nonzero rate even for subthreshold mean drive.

intersection (V^*, w^*) is the rest state. *Linearizing*, compute the Jacobian $\mathbf{J} = \partial(\dot V, \dot w)/\partial(V, w)$ entry by entry:

$$\mathbf{J} = \begin{pmatrix} \dfrac{1}{C_m}\dfrac{\partial F}{\partial V} & \dfrac{1}{C_m}\dfrac{\partial F}{\partial w} \\ \dfrac{1}{\tau_w}\dfrac{\partial G}{\partial V} & \dfrac{1}{\tau_w}\dfrac{\partial G}{\partial w} \end{pmatrix} = \begin{pmatrix} \dfrac{g_L\left(e^{(V^*-V_T)/\Delta_T} - 1\right)}{C_m} & -\dfrac{1}{C_m} \\ \dfrac{a}{\tau_w} & -\dfrac{1}{\tau_w} \end{pmatrix}. \tag{30}$$

Stability is read from $\operatorname{tr}\mathbf{J}$ and $\det\mathbf{J}$: a **saddle-node** bifurcation ($\det\mathbf{J} = 0$) gives Type-I onset and regular firing, while an **Andronov–Hopf** bifurcation ($\operatorname{tr}\mathbf{J} = 0$, $\det\mathbf{J} > 0$) gives Type-II onset, subthreshold resonance, and bursting. Tuning (a, b, τ_w, V_r) reproduces tonic, adapting, bursting, and irregular firing.

7.4 Spike-frequency adaptation, in detail

Many neurons fire rapidly at stimulus onset and then slow down — *spike-frequency adaptation.* The minimal model augments the LIF with a slow adaptation current w:

$$\tau_m \dot V = -(V - E_L) + RI - Rw, \tag{31}$$

$$\tau_w \dot w = -w, \qquad \text{with } w \to w + b \text{ at each spike}, \quad \tau_w \gg \tau_m. \tag{32}$$

Each spike increments w by b; between spikes w decays with the slow constant τ_w. *Adapted rate.* Under steady firing at rate r, average $\tau_w \dot w = -w$ over one inter-spike interval: the spike adds b per interval $1/r$, i.e. a mean input rate br to w, balanced by the decay $\bar w/\tau_w$, so

$$\frac{\bar w}{\tau_w} = b\,r \quad\Longrightarrow\quad \bar w \approx b\,\tau_w\,r. \tag{33}$$

This $\bar w$ feeds back negatively: the effective drive becomes $RI - R\bar w = RI - Rb\tau_w r$, so the adapted rate solves the self-consistency $r = \Phi_{\mathrm{LIF}}(RI - Rb\tau_w r)$ and lies *below* the onset rate $\Phi_{\mathrm{LIF}}(RI)$. The inter-spike intervals therefore lengthen from an initial T_0 toward a longer steady value (Fig. 12): b sets the *strength* of adaptation and τ_w its *timescale.* Functionally this is high-pass temporal filtering — the neuron responds strongly to input *transients* and weakly to sustained input.

Which firing regimes can be reproduced? A reset map plus a single slow variable is already enough to generate most cortical firing patterns, selected by the strength/timescale of the slow current and by the reset values:

- **tonic spiking** — weak/no adaptation ($b \to 0$): regular, clock-like firing;

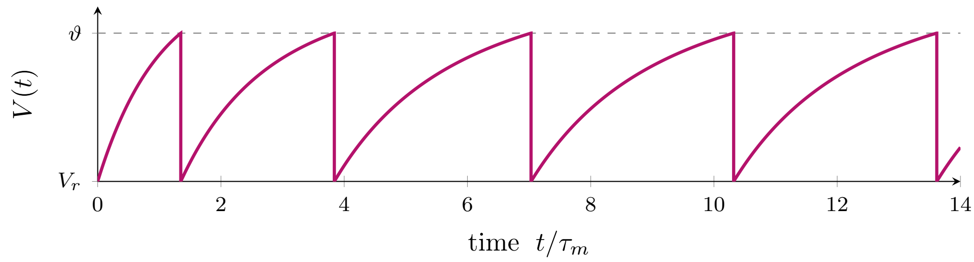

Figure 12: Spike-frequency adaptation (simulated LIF with a slow adaptation current). The first interval is short; the spike-triggered current w accumulates and progressively lengthens the intervals until they settle — the signature of an adapting neuron.

- **spike-frequency adaptation** — moderate b, slow τ_w: intervals lengthen then settle (Fig. 12);
- **initial (onset) bursting** — a brief high-frequency burst at onset, then adapted single spikes;
- **regular/tonic bursting** — a depolarized reset together with the slow current yields repeated bursts;
- **irregular firing** — adding noise (Euler–Maruyama) to any of the above gives $C_V \approx 1$.

The AdEx and Izhikevich models organize these regimes explicitly through their two-variable phase planes; we now study the Izhikevich model in detail.

7.5 The Izhikevich model, in detail

Izhikevich (2003) distilled the spike/recovery interplay into two equations that reproduce essentially every cortical firing class at the cost of a few operations per step. In dimensional form (v in mV, t in ms),

$$\dot v = 0.04\,v^2 + 5v + 140 - u + I, \qquad \dot u = a\,(b\,v - u), \qquad v \ge 30 \Rightarrow \begin{cases} v \leftarrow c, \\ u \leftarrow u + d. \end{cases} \tag{34}$$

Here v is the membrane potential and u a slow recovery variable (lumping K^+ activation and Na^+ inactivation). The quadratic $0.04v^2 + 5v + 140$ is the quadratic-IF spike nonlinearity, calibrated so that v is in mV with rest near -70 mV and threshold near -55 mV; the cutoff at $+30$ mV is bookkeeping — once reached, v is reset to c and u stepped up by d. The four parameters have clear meanings:

- a — the *rate* of recovery ($1/a$ is its time constant): small a = slow recovery = stronger adaptation and bursting;
- b — the *subthreshold coupling* of u to v: larger b makes the pair resonant (sag, rebound spikes, subthreshold oscillations);
- c — the post-spike *reset* of v: a more depolarized c favours bursting;
- d — the post-spike *jump* in u: larger d = stronger spike-triggered adaptation.

Varying only (a, b, c, d) selects the regime (Fig. 13). Canonical cortical settings: *regular spiking* RS $(0.02, 0.2, -65, 8)$; *intrinsically bursting* IB $(0.02, 0.2, -55, 4)$; *chattering* CH $(0.02, 0.2, -50, 2)$; *fast spiking* FS $(0.1, 0.2, -65, 2)$; *low-threshold spiking* LTS $(0.02, 0.25, -65, 2)$. The model is integrated by the same explicit scheme as before; because the v^2 term grows quickly one uses a small step (e.g. two half-steps of 0.25 ms) together with the $+30$ mV cutoff.

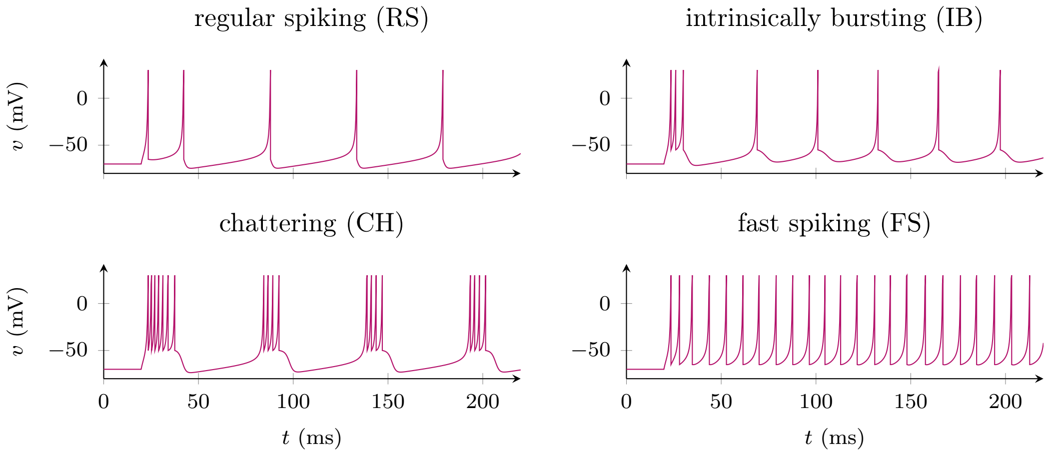

Figure 13: Firing regimes of the Izhikevich model (34) under a step current, obtained by varying only (a, b, c, d): regular spiking with adaptation (RS), intrinsic bursting (IB), fast rhythmic bursting / chattering (CH), and fast spiking (FS). Simulated with the two-half-step explicit scheme. After [8].

7.6 Spike-response model, GLM, and conductance-based input

The **spike-response model** $V(t) = \sum_{t_f} \eta(t - t_f) + (\kappa * I)(t) + V_{\text{rest}}$ with a stochastic escape rate $\lambda = f(V - \vartheta)$ becomes the convex-likelihood **GLM** fit routinely to data [1]. **Conductance-based** input $I_{\text{syn}} = \sum_x g_x(t)(E_x - V)$ shortens the effective time constant $\tau_{\text{eff}} = C_m/(g_L + \sum_x \bar{g}_x)$ (high-conductance state).

8 Synaptic input: conductances and filtering

A presynaptic spike at time t_f opens postsynaptic channels, producing a transient conductance $g_{\text{syn}}(t) = \bar{g} \sum_f s(t - t_f)$ with reversal potential E_{syn}, so the synaptic current is $I_{\text{syn}} = g_{\text{syn}}(t)(E_{\text{syn}} - V)$. Two standard kernels:

- **Exponential synapse**: $s(t) = e^{-t/\tau_s}\Theta(t)$, the impulse response of $\tau_s \dot{s} = -s$ kicked by $+1$ at each input;
- **Alpha synapse**: $s(t) = \frac{t}{\tau_s} e^{-t/\tau_s}\Theta(t)$, with a finite rise time (two cascaded exponential filters).

In the *current-based* approximation ($E_{\text{syn}} - V$ roughly constant) the input is a filtered spike train, and by (17) the voltage is a *double convolution* $V - E_L = \kappa * s * \rho$ — the membrane filter κ in series with the synaptic filter s. For an exponential synapse the postsynaptic potential is obtained by convolving κ with s:

$$(\kappa * s)(t) = \frac{1}{C_m} \int_0^t e^{-(t-u)/\tau_m} e^{-u/\tau_s} \, \mathrm{d}u = \frac{\tau_m \tau_s}{C_m(\tau_m - \tau_s)} \left(e^{-t/\tau_m} - e^{-t/\tau_s}\right), \tag{35}$$

a difference of exponentials that rises on the fast timescale and decays on the slow one. In the *conductance-based* case $g_{\text{syn}}V$ is multiplicative and shortens the effective time constant, as in the high-conductance state.

9 Spike-train variability: CV and Fano factor

Two measures quantify irregularity. The **coefficient of variation** of the inter-spike intervals, $C_V = \sigma_{\text{ISI}}/\langle\text{ISI}\rangle$, is 0 for a clock-like train and 1 for a Poisson process. The **Fano factor** of the spike count N in a window W, $F = \text{Var}[N]/\langle N\rangle$, is 1 for Poisson and $\to 0$ for a regular train; for a renewal process $F \to C_V^2$ at long W. *Poisson derivation*: if spikes are Poisson at rate r, the count in W is Poisson(rW) so $\mathrm{Var}[N] = \langle N \rangle = rW \Rightarrow F = 1$, while the ISIs are exponential with mean and standard deviation both $1/r \Rightarrow C_V = 1$. The deterministic LIF gives $C_V = F = 0$; the fluctuation-driven noisy LIF (Fig. 10) interpolates to $C_V \approx 1$, matching cortical variability.

10 A worked numerical example

Take $\tau_m = 20\,\mathrm{ms}$, $R = 100\,\mathrm{M\Omega}$, $E_L = -70\,\mathrm{mV}$, $\vartheta = -50\,\mathrm{mV}$, $V_r = -70\,\mathrm{mV}$, $\tau_{\mathrm{ref}} = 2\,\mathrm{ms}$.

(a) Rheobase.
$I_{\mathrm{rh}} = (\vartheta - E_L)/R = (20\,\mathrm{mV})/(100\,\mathrm{M\Omega}) = 0.2\,\mathrm{nA}$.

(b) Rate at $I_0 = 0.4\,\mathrm{nA}$.
The asymptote is $E_L + RI_0 = -70 + 40 = -30\,\mathrm{mV}$, so

$$T = \tau_m \ln \frac{E_L + RI_0 - V_r}{E_L + RI_0 - \vartheta} = 20\,\mathrm{ms}\,\ln \frac{-30+70}{-30+50} = 20 \ln \tfrac{40}{20} = 20 \ln 2 \approx 13.9\,\mathrm{ms},$$

hence $r = 1/(\tau_{\mathrm{ref}} + T) = 1/(15.9\,\mathrm{ms}) \approx 63\,\mathrm{Hz}$.

(c) Doubling the current
to $0.8\,\mathrm{nA}$ raises the asymptote to $+10\,\mathrm{mV}$; $T = 20 \ln \frac{80}{60} = 20 \ln \frac{4}{3} \approx 5.75\,\mathrm{ms}$ and $r \approx 1/(7.75\,\mathrm{ms}) \approx 129\,\mathrm{Hz}$ — roughly doubling, consistent with the near-linear strong-drive regime.

Exercises

1. Show from (19) that the 10–90% rise time of the membrane to a current step is $\tau_m \ln 9 \approx 2.2\,\tau_m$.
2. Derive the EPSP (difference of exponentials) by convolving κ with an exponential synapse, and find the time of its peak.
3. Using (23), show that the f–I onset is continuous (Type-I) and that for $\tau_{\mathrm{ref}} > 0$ the rate saturates at $1/\tau_{\mathrm{ref}}$.
4. Compute $|Z(\omega)|/R$ and the phase lag at $\omega\tau_m = 1$.
5. (Numerical) Implement the Euler–Maruyama scheme (26) for (25); measure the f–I curve and C_V in the mean- and fluctuation-driven regimes, and check that the results are insensitive to halving Δt.
6. (Numerical) Simulate the Izhikevich model (34) and reproduce the regular-spiking, bursting, and fast-spiking regimes by varying (a, b, c, d).

11 Summary

We reduced a spiking cell to the LIF filter (17) plus a reset, deriving every step: the Nernst potential (4), the membrane equation (9), the integrating-factor solution (14), the Dirac Green's function (16), the harmonic impedance (21), the f–I inversion (23), and the noisy neuron simulated by the Euler–Maruyama scheme (26). The nonlinear and adaptive extensions — spike-frequency adaptation, AdEx, and the Izhikevich model — add spike shape, adaptation, and the full diversity of firing patterns while remaining computationally light [1, 8].

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